

Topic: 1

Name: Telemedicine & Coverage Checker

Description: App to schedule virtual care and verify if a health plan covers it.

- **Features:**
 - Video integration (e.g., Twilio)
 - Insurance eligibility check
 - Doctor profiles
- **Problem:** Not all plans cover virtual care. Users need clarity.
- **Key Features:**
 - Search for telehealth doctors
 - Real-time appointment booking
 - Check insurance coverage by plan
- **APIs:**
 - Twilio Video API (telemedicine)
 - Firebase Auth
 - Simulated plan benefit data
- **Real-World Use:**
 - Like Teladoc or MDLive, but focused on insurance validation.
- **Overview:** Book telehealth and verify coverage.
- **Architecture:**
 - Mobile-first
 - Real-time video sessions
- **Modules:**
 - Appointment Booking
 - Insurance Check
 - Video Chat

Topic: 2

Name: Billing & Premium Payment Portal

Description: Build a system for members to view and pay their monthly premiums securely.

- **Features:**
 - Invoices and receipts
 - Auto-pay setup
 - Secure payment integration
- **Problem:** Insurers need secure, user-friendly portals to collect premiums.
- **Key Features:**
 - Payment history
 - Stripe/PayPal integration
 - Auto-pay, reminders
- **APIs:**
 - Stripe Billing API
 - Email/SMS (Twilio)
- **Real-World Use:**
 - Core function in any private or ACA-based plan provider.
- **Overview:** View and pay insurance bills online.
- **Architecture:**
 - Stripe Webhooks
 - Secure payment endpoints
- **Modules:**
 - Billing History
 - Payment Page
 - Auto-pay Settings

Topic: 3

Name: Provider Finder & Network Lookup

Description: App for users to check which doctors or hospitals are in-network for a given insurance plan.

- **Features:**
 - Search/filter by provider specialty, location, insurance
 - Map integration
 - Reviews and rating system
- **Problem:** Patients often go out-of-network unintentionally, leading to high costs.
- **Key Features:**
 - Search doctors by specialty, insurance, zip
 - Map view with distance filtering
 - Doctor profiles and ratings
- **Possible APIs:**
 - NPI Registry API
 - Google Maps API
 - ZocDoc API (unofficial scraping may be required)
- **Real-World Use:**
 - Used by insurance websites like Aetna or UHC to help users find covered doctors.
- **Overview:** Lets users find in-network doctors by location, specialty, and insurance.
- **Architecture:**
 - SPA with search queries
 - API Caching for performance
- **Modules:**
 - Search Interface
 - Doctor Profile Viewer
 - Google Maps Integration
- **Database:**
 - Provider {
 - id,
 - name,
 - specialty,
 - address,
 - insuranceList }
- **Security:**
 - Input sanitization
 - Geo-query throttling

Topic 4

Name: Admit, Room Transfer & Discharge System

"Trilife hospital" needs to automate the patient admission, discharge & transfer process (Room Transfer) which is currently depending on huge manual interventions. Based on their customer feedbacks , they want to build a web application which eases these process. The complete system should be automated including the insurance clearance which is raised by the patient. This module will support only cash less based services provided by the insurance agencies

Functional Requirements

1. Patient

- Able to view the availability of Rooms
- Book for a specific room / Request for Transfer the rooms
- Get the status of request
- Notifications on Discharge
- Follow procedure of cashless services

2. Admin

- Access all the patient request
- Approve or cancel the room request with proper response
- Should able to view the available room details
- Generate discharge details
- Communicate with Discharge Team

3. Front Desk

- Patient admission
- Maintains all discharge details
- Verifies the clearance of insurance claim for cashless services

Topic 5

Name: Health Checkup For Employees

The company named “**WeSuccess**” wants to automate their process of health check up for their employees. The company has tie up with insurance agency and they wanted their employees to get the benefit with simple and supportive transactions online. The insurance agency provides list of their preferred network hospitals online with procedures to be followed up for getting the appointment online. The details of the tests covered, and names of the enrolled hospitals should be available in the mobile app. The process to be followed for booking appointments to be followed up by the employees

Functional Requirements

1. Company

- Build partnership with insurance agency
- Set up policy for eligibility
- Verifies the utilization of this scheme yearly
- Provide facilities for the employees to utilize this scheme

2. Insurance Agency

- Provides the details of preferred partner hospital
- Verifies the employee booking details provided by the hospital admin
- Approve or reject the request based on company policy on eligibility

3. Employee

- Need to book appointment online with preferred partner
- Gets notification about the status of his bookings
- Reschedule or cancel the request
- View his health records online

4. Hospital Admin

- Sends the details with insurance agency for verification
- Updates the health record once the test are completed

Topic 6

Title: Pharmacy Management System

HealthSure Hospital has its own in house pharmacy, which needs a Software to manage their transactions based on the requirements. The wanted to automate the process of managing their inventory in warehouse and keep a check on the medicines sold and medicines in stock. They need to raise Invoices to the Retailers and also billing to the patients who purchase medicines. This automated system should support them in properly delivering the medicines on time for the patients without making them to wait for longer period

Functional Requirements

1. Patient:

- Able to register themselves
- Submit request for medicines online
- Track the status of delivery
- Access bills
- Customer can also write there feedback about the service provided by pharmacy

2. Pharmacy Admin

- Check availability of medicines and other pharmacy products
- Maintains the stock
- Place order to retailers based on stock availability and requirements
- Generate reports
- Provide any Schemes for customers

3. Pharma Incharge

- Deliver the medicines
- Update the stock details
- View the details of available medicines
- Generate bills

Topic 7

Title: Outpatient Schedule Tracking

"Sarvotham's Spine Care" is one of the growing hospital at Bengaluru. They are getting more popular due to their services and witnessing growth in out patients day to day. They have effective people support system, however all transactions are maintained in manual register. They want to build a digital solution that allows the hospital to manage and track every out patient's schedule. It will enhance the productivity of the hospital and allow the attending physicians and nurses to better manage the patients.

Functional Requirements

1. Front Desk Kiosk:

- Able to align out patient with specific doctor
- Track the status of patient
- Redirect the request to specific support staff

2. Support Staff (Nurse / Attender / Lab Technician)

- Update the health issues of patient online
- Collects the diagnostic reports and update patient profile
- Doctor treatment fees and any additional cost to be provided

3. Patient

- Should able to register
- View his treatment records
- Able to access his bills and lab reports

4. Billing Desk

- Verifies the treatment type
- Provide bills based on all treatments
- Process billing with / without cash based on claims

Topic 8

Title: Practo App

We want to build a software like **Practo App** to automate doctor appointment system for customers across the group of hospitals. The patient should be able to view the doctors details based on their geo location and book an appointment online. The doctor should be able to access and approve or deny the request. The hospital admin should be able to schedule an appointment and track the record. We need to develop a web application to support customers without walking in or making calls to the hospitals.

Functional Requirements

1. Patient

- Able to explore the doctor details based on their health issues and geo location
- Should be able to request for rescheduling of appointment
- View the status of his request
- Able to connect with hospital admin for the processing

2. Doctor

- To fill their availability
- Responsible for approving or rejecting the appointment
- Once treatment is over , sends the confirmation to hospital admin
- Updates Patient records

3. Hospital Admin

- Sends the request of Patients to Doctor based on their availability
- Fixes the appointment and share the details with Patient
- Follow up the appointment and maintains details.

Topic 9

Title: Health Insurance Application

Niva Bupa Life insurance Ltd is an insurance company and they want to make Online Insurance Portal. Currently they are getting multiple request in person and through email from their partner companies and hospitals. Niva Bupa wants to sell different types of policies to customer who needs to purchase it online.

Functional Requirements

1. Customer

- Explore what Nivabupa's wide range of policies
- Select and make a purchase of a particular policies
- Upload necessary documents [PAN and Aadhar]
- Portal to manage all Niva bupa policies in one roof

2. Life Insurance Company

- Bring new Policies into the Market
- Approve a Policy to a Customer
- Message Communications with Customer
- Emailing Policy documents to the registered Customers
- Renewal Process

Topic 10

Title: Medication-Management-Tool

Managing different medications can be pretty cumbersome for a patient, especially if they are taking medication for different ailments.

First, there's the issue of keeping the schedule of every medication in mind. Then, they have to manage the quantity of every medication and get more before one of them runs out. Missing on dosage can be very harmful to a patient's well-being.

We need to create a digital solution for helping people with medication management. They would have to enter their different medication and its schedule once into the application. Then the application will alert them about their dosage time and track how many doses they have taken to remind them about buying more medicine.

A personal management tool that allows users to input prescription details, and send reminders for when they need to be taken or refilled. This tool is ideal for users with multiple different prescriptions to keep track of and aims to help reduce the likelihood of missing doses.

Functional Requirements

1. Develop a system for tracking prescriptions and dosages.
2. Include features for alerts and reminders for both patients and providers.
3. Integrate with pharmacy systems for real-time updates
4. Input prescription details, including medication name, dosage, and instructions.
5. Set customizable medication reminder notifications.
6. Receive notifications for upcoming medication doses.
7. Get alerted when it's time to refill your prescriptions.
8. Store important information about your healthcare providers and pharmacies.

Topic 11

Title: Staffing Management System

Hospitals and clinics have to rely on staffing agencies to find and hire nurses and other staff. In most cases, the staffing agency doesn't find all the required professionals in time and handle the process very inefficiently. The lack of staff can be very detrimental to a hospital's proper functioning. The solution would connect hospitals and clinics with skilled and verified candidates. By getting rid of the middleman, you can save both parties a ton of time.

Functional Requirements

1. Hospital Staff Allocation Team

- Login
- Search for required Staff from the portal
- Should be able to request for staff allocation [For a Particular Date and Time]
- Management of Staff

2. Nurse / Care Provider

- Login
- To fill their availability
- Responsible for approving or rejecting the request
- Updates & Maintain Patient records
- View the attendance report of themselves

3. Admin

- Login
- Register new employees to the system
- View attendance report of all employees.
- Attendance can be filtered by date or employee.

Topic 12

Title: Medical Equipment Procurement & Management System

Hospitals need to procure Medical Equipment's, Medical Equipment Procurement Management System (MEPMS) are deployed in healthcare facilities to monitor, and manage medical equipment's. A fully functional MEPMS system is needed for the provision of optimal healthcare services in hospitals. A medical equipment inventory provides detailed records of medical equipment within the healthcare facility. It indicates the level of functionality of each device in order for support and technical staff to take appropriate measures in fixing faulty medical equipment. In recent years, the rapid improvement of clinical technology has led to a significant increase in the number and variety of medical equipment produced for diagnostic and therapeutic purposes. Thus, proper inventory management system for medical equipment and deployment of a suitable maintenance strategy are imperative. How to manage and preserve these medical devices of diverse brands is a major problem for hospitals. In medium, large and very large hospital settings, the volume of medical equipment used to provide healthcare services could be heterogeneous and very complicated resulting to increased need for equipment preservation management. Improved usability and reliability of most medical equipment and spares are guaranteed with significant reduction in wastage if a system of proper inventory and maintenance management is set up.

Inventory management can be performed on a web based system. Inventory management is essential to enhanced administration of healthcare technology. The inventory system should maintain an up-to-date record consistently so that at any given moment, it provides the right information about the number of functional medical equipment in the healthcare facility.

The biomedical or clinical engineering department in any healthcare system is responsible for the effective workflow of medical equipment in the health technology management system including tests, repairs and maintenance thereby ensuring a safe and effective healthcare service provision.

The development of a web based (MEPMS) software, will be the most effective approach to manage hospital equipment where the hospital, health professionals, patients, engineering unit, and other researchers could leverage from the platform to improve their work proficiency and provide safe health care delivery to patients.

Functional Requirements

The aim of this study is to develop a web-based medical equipment inventory management system. The specific objectives of the study are to:

1. Identify key requirements for the design of a medical equipment inventory management system
2. Develop core modules of the medical equipment inventory management system.

3. Integrate all the modules into a system
4. It Should presents a summary of the total number of equipment
5. The total registered users
6. The total medical equipment under repair
7. The total equipment purchased
8. Monitoring log
9. Usage analysis
10. Provides further information on the newly added equipment to the MEPMS indicating the type of equipment, quantity, and their category whether diagnostic or therapeutic.